1.Define Power BI and What are the key components of the Power BI ecosystem? Briefly explain:
● Power BI Desktop
● Power BI Service
● Power BI Mobile
● Power BI Gateway

Power BI is a Microsoft business analytics suite that helps users turn raw data into actionable, interactive visualizations. Its key components include Power BI Desktop for report creation, the Power BI Service for sharing and collaboration, Power BI Mobile for viewing on the go, and Power BI Gateway for connecting on-premises data sources to the cloud.

Key Components

- Power BI Desktop: A free Windows application used to connect to data, model it, and create interactive reports with visuals like charts and maps. You use it to build the initial reports by cleaning, shaping, and analyzing data from various sources.
- Power BI Service: A cloud-based platform where you can publish reports created in Power BI Desktop and then share them with others. It also allows for the creation of dashboards and provides features for collaboration and consuming insights from a web browser.
- Power BI Mobile: Mobile applications available for iOS, Android, and Windows devices that allow users to access, view, and interact with reports and dashboards created in the Power BI Service while on the move.
- Power BI Gateway: A service that acts as a secure bridge, allowing the Power BI Service to connect to and refresh data from on-premises data sources, such as local databases or files. This is crucial for organizations that keep their data within their own network.

2. Compare the following Power BI visuals:

● Pie Chart vs Donut Chart

● Bar Chart vs Column Chart When would you prefer one over the other? Give one example for each pair.

A donut chart is preferred over a pie chart when you need to display a label or total value in the center, making it suitable for a smaller number of categories, whereas a pie chart is better for quickly showing a part-to-whole relationship with a single data series. A bar chart (horizontal) is best for a large number of categories or long labels, while a column chart (vertical) is ideal for a smaller number of categories and comparing values that are close in size.

Pie Chart vs. Donut Chart

Feature	**Pie Chart**	**Donut Chart**
Core Function	Shows the proportion of parts to a whole.	Shows the proportion of parts to a whole, with a hollow center.
Key Difference	A filled-in circle.	A hole in the center that can display a label, icon, or total value.
Best For	Quick, at-a-glance comparison of a few categories, like market share, where the center isn't needed for extra information.	Displaying a key value (like a total) in the center while showing its components. Also useful for interactivity, where the donut can act as a filter for other visuals.

When to Use	When you need to show a part-to-whole relationship with a single data series and the segments are easy to distinguish.	When you need to add a label or value in the center or want a more modern-looking chart.
Example	Showing the percentage breakdown of a company's revenue by major product line (e.g., 4-5 categories).	Displaying a company's overall sales total in the center, with each arc representing the sales contribution of a different region.

Bar Chart vs. Column Chart

Feature	**Bar Chart**	**Column Chart**
Core Function	Compares different categories using horizontal bars.	Compares different categories using vertical bars.
Key Difference	Horizontal orientation.	Vertical orientation.
Best For	When you have many categories to compare or category labels are long and would overlap on a vertical axis.	When you have a smaller number of categories to compare or when comparing values that are close to each other, as the vertical axis makes it easier to compare height.

When to Use	Ideal for long lists of items, such as a product comparison or a list of countries by population, where the horizontal space is beneficial.	Good for time-series data or when a vertical comparison is more natural, such as comparing monthly sales figures or website visitors per day.
Example	Comparing the market share of 20 different companies where each company name is a long string of text.	Comparing the quarterly sales performance of your top four products.

3. Explain the significance of:

● Star schema vs Snowflake schema

● Primary key vs Foreign key in relationships (Power BI) Why is cardinality important?

Star Schema:

- Characterized by a central fact table surrounded by denormalized dimension tables, forming a star-like structure.
- **Significance:** Simplicity, faster query performance due to fewer joins, and ease of understanding for business users. Ideal for ad-hoc reporting and scenarios where quick aggregation is prioritized.

Snowflake Schema:

- An extension of the star schema where dimension tables are further normalized into multiple related tables, resembling a snowflake.
- **Significance:** Reduced data redundancy, improved data integrity, and better storage efficiency. Suitable for complex analytical needs and large data warehouses requiring detailed dimensional hierarchies.

Primary Key vs. Foreign Key in Relationships (Power BI)

Primary Key:

- A column or set of columns in a table that uniquely identifies each row in that table.
- **Significance:** Ensures data integrity by guaranteeing uniqueness and non-null values for each record, enabling efficient retrieval and referencing of specific rows.

Foreign Key:

- A column or set of columns in one table that refers to the primary key in another table, establishing a link between the two.
- **Significance:** Enforces referential integrity between tables, meaning that a foreign key value must either be null or match an existing primary key value in the referenced table. This maintains consistency and allows for the creation of meaningful relationships for data analysis.

Importance of Cardinality

Cardinality:

- Describes the nature of the relationship between two tables, specifically how many instances of one entity are related to how many instances of another entity. Common types include one-to-one, one-to-many, and many-to-many.
- **Significance:**

- **Data Modeling:** Crucial for designing effective data models in Power BI, as it dictates how tables are joined and how data flows between them.
- **Query Performance:** Incorrect cardinality can lead to inaccurate results or performance issues in Power BI reports and calculations.
- **DAX Calculations:** Understanding cardinality is fundamental for writing accurate DAX formulas, especially when dealing with aggregation and filtering across related tables.
- **Relationship Integrity:** Ensures that relationships between tables are correctly defined, preventing data inconsistencies and enabling reliable analysis.

4. Differentiate between:

● Calculated column vs Measure

Also, define Row context and Filter context with simple examples.

Calculated Column:

- **Definition:** A new column added to an existing table in your data model, where each cell's value is calculated row by row using a DAX formula.
- **Evaluation:** Calculated at the time of data refresh or when the column is initially created. Stored in the data model, consuming memory.
- **Usage:** Used to add new attributes or categories to your data, or when the calculation is independent of user interaction in reports (e.g., calculating age from birthdate). Can be used in slicers, rows/columns of pivot tables, or as filters.
- **Example:**

Code

```
Age = DATEDIFF('Customers'[BirthDate], TODAY(), YEAR)
```

Measure:

- **Definition:** A dynamic calculation that aggregates data based on the current filter context in a report.
- **Evaluation:** Calculated on-the-fly when used in a visual or report, not stored in the data model. Consumes CPU during calculation.
- **Usage:** Used for aggregations and calculations that depend on user selections and filters (e.g., total sales, average profit margin). Cannot be used directly in slicers or as row/column headers in pivot tables.
- **Example:**

Code

```
Total Sales = SUM('Sales'[Sales Amount])
```

Row Context and Filter Context

Row Context:

- **Definition:** Refers to the "current row" in a table during a row-by-row calculation. When a DAX expression is evaluated within a row context, it has access to the values in all columns of that specific row.
- **Implication:** Implicitly present when creating calculated columns. Each row is processed individually.
- **Simple Example:** Consider a calculated column `Profit` in a `Sales` table with `Revenue` and `Cost` columns.

Code

```
Profit = 'Sales'[Revenue] - 'Sales'[Cost]
```

For each row, the formula calculates the profit using the `Revenue` and `Cost` values from that specific row.

Filter Context:

- **Definition:** The set of filters applied to the data model by visuals, slicers, report-level filters, or other DAX functions. These filters narrow down the data set on which a calculation is performed.
- **Implication:** Measures are always evaluated within a filter context.
- **Simple Example:** Consider a measure `Total Sales` in a `Sales` table.

Code

```
Total Sales = SUM('Sales'[Sales Amount])
```

If you place this measure in a visual and filter the report by `Region = "North"`, the `Total Sales` measure will only sum the `Sales Amount` for rows where the `Region` is "North." The filter context here is `Region = "North"`.

5. What is the difference between a report and a dashboard in Power BI?

A Power BI report provides a detailed, multi-page view of data from a single data model, allowing for in-depth analysis and interactive exploration. A dashboard, by contrast, is a one-page, high-level summary of key metrics often pulled from multiple reports and datasets, designed for a quick glance at performance. Dashboards are less interactive than reports, which feature cross-filtering and drill-down capabilities.

Feature	Power BI Report	Power BI Dashboard
Primary Purpose	Detailed, in-depth data analysis and exploration	High-level, quick monitoring of key metrics

Data Sources	Typically from a single data model	Can combine visuals from multiple reports and datasets
Number of Pages	Can have multiple pages of visuals	Always a single page
Interactivity	Highly interactive, with cross-filtering and drill-down capabilities	Limited interactivity; visuals are static with filters pre-applied
Customization	Highly customizable with multiple pages and filters	Limited customization; visuals can be rearranged and resized, but not heavily modified
Use Case	For deep dives into data, finding patterns, and detailed reporting	For real-time monitoring of KPIs to make quick decisions

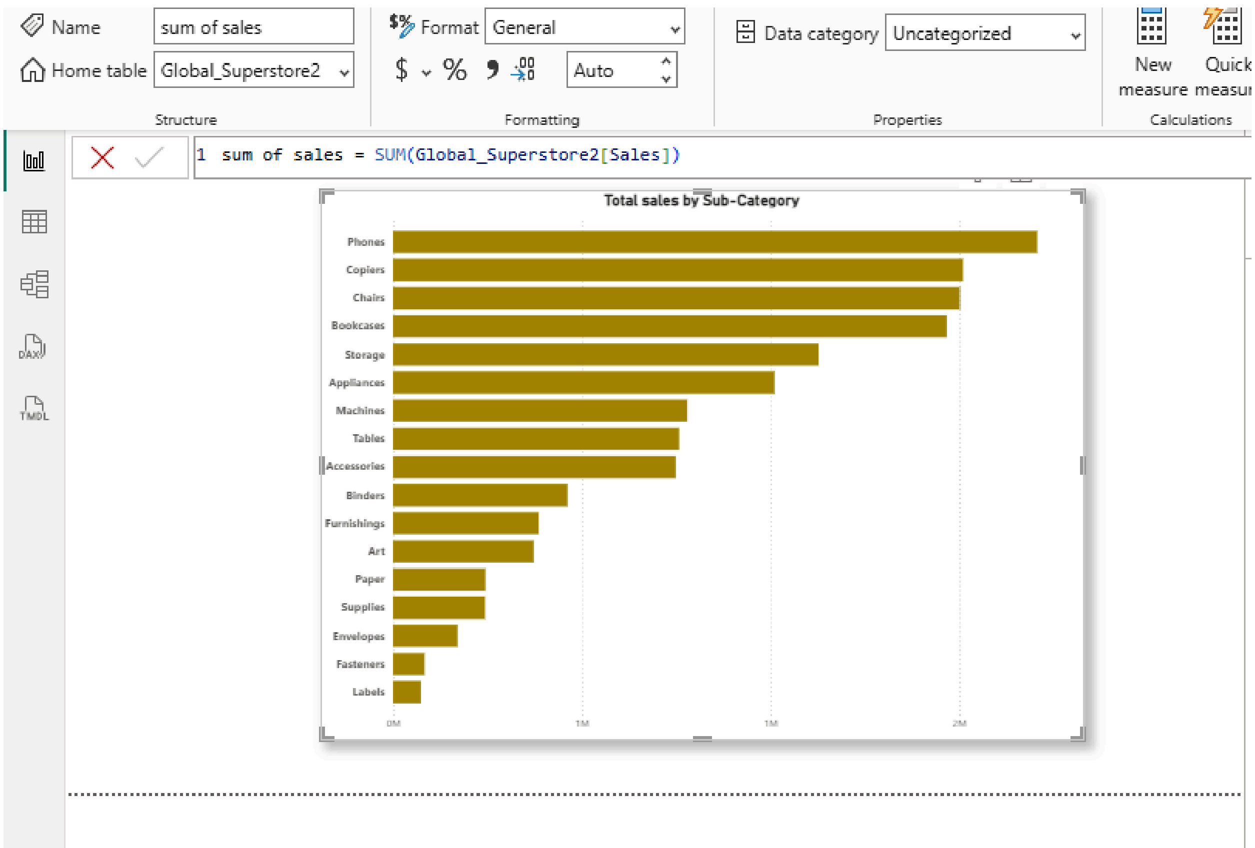
When to use each:

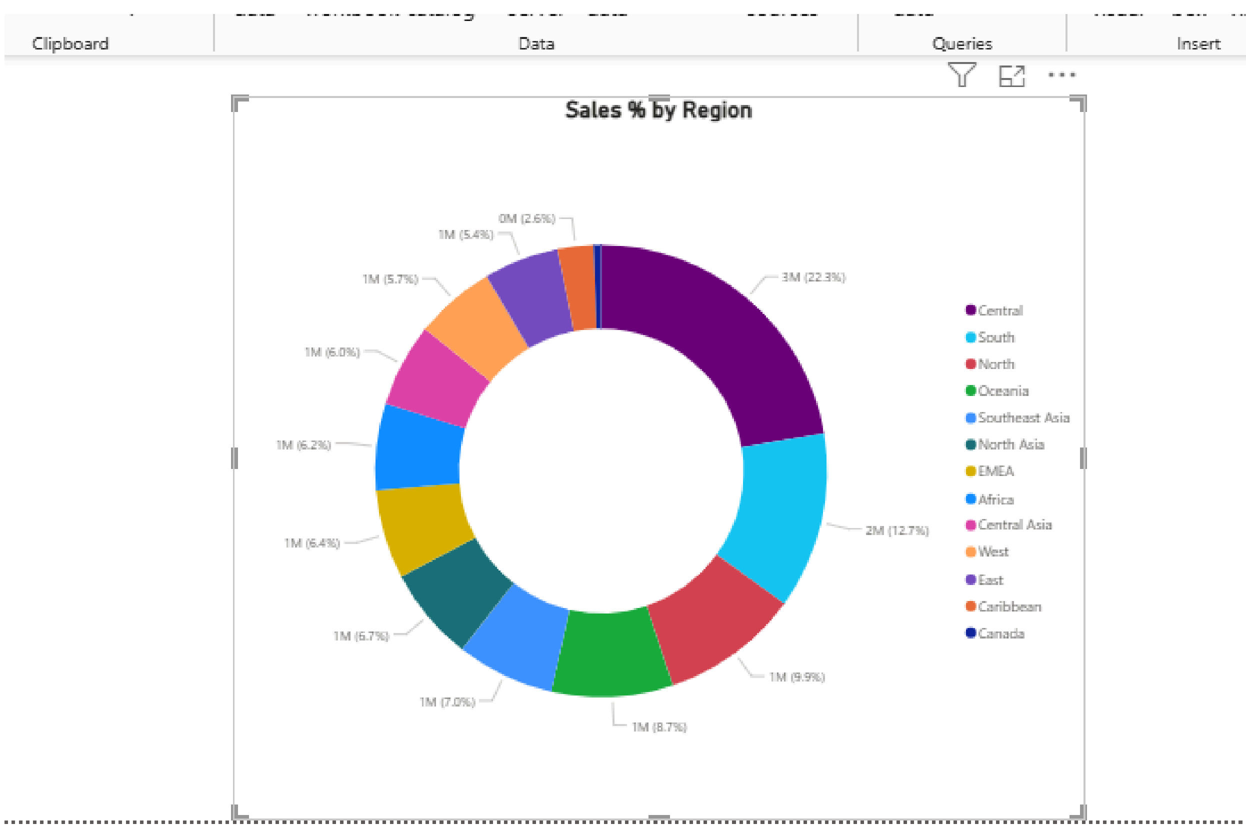
- **Use a report** when you need to explore data in detail, uncover trends, and provide a comprehensive analysis of a specific topic or dataset.
- **Use a dashboard** when you need to see a quick, consolidated view of your most important metrics from different areas to monitor performance at a glance

6. Using the Sample Superstore dataset:

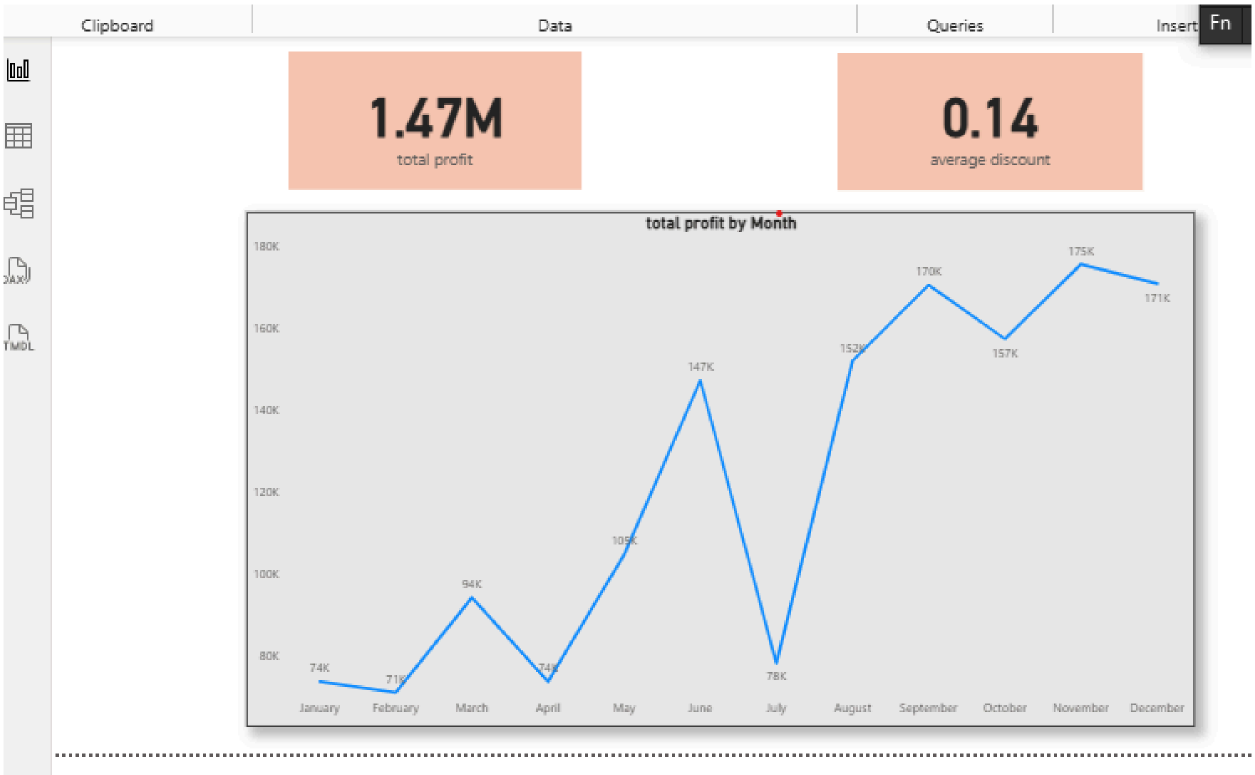
● Create a Clustered Bar Chart to display Total Sales by Sub-Category

- Create a Donut Chart for Sales % by Region Provide screenshots of both visuals.





7.Write and apply the following measures: ● Total Profit = SUM([Profit]) ● Average Discount = AVERAGE([Discount]) Display both in a KPI Card, and use a Line Chart to show profit trend over months. Add visuals and DAX formulas.



8. Implement a DAX measure that calculates the percentage of total sales by product category.

Product_category	Sales_Amount
Electronics	5000
Clothing	3000
Home Appliances	7000
Books	2000
Tables & Chairs	8000
Toy	1500
Sports Equipment	1200
Office Supplies	1000
Beauty Products	4400
Garden Supplies	1000
Jewelry	1800
Automative	2600

```
% of Total Sales =
DIVIDE(
    SUM('SalesTable'[Sales_Amount]),
    CALCULATE(SUM('SalesTable'[Sales_Amount]), ALL('SalesTable'))
)
```

Explanation:

- `SUM('SalesTable'[Sales_Amount])`: Calculates the total sales for each row/context (i.e., each product category).
- `CALCULATE(SUM(...), ALL('SalesTable'))`: Gets the **grand total** of sales across **all** categories.
- `DIVIDE(...)`: Safely divides the category total by the grand total to avoid divide-by-zero errors.

💡 Result Example (based on your data):

Product Category	Sales Amount	% of Total Sales
Electronics	5000	11.36%
Clothing	3000	6.82%
Home Appliances	7000	15.91%
Books	2000	4.55%
Tables & Chairs	8000	18.18%
Toy	1500	3.41%
Sports Equipment	1200	2.73%
Office Supplies	1000	2.27%
Beauty Products	4400	10.00%
Garden Supplies	1000	2.27%
Jewelry	1800	4.09%
Automative	2600	5.91%

Total	**44000**	**100%**

9. Create a DAX Measure for Total Profit ● Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit ● Add a Slicer for Region to filter the visual ● Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit. Provide a steps, screenshot of the Waterfall chart and the DAX formula

Step 1: Create a DAX Measure for Total Profit

Assuming you have a dataset with `Sales`, `Cost`, and `Sub-Category`, use this DAX formula:

```
Total Profit = SUM('SalesData'[Sales]) - SUM('SalesData'[Cost])
```

> Replace `'SalesData'` with the actual name of your table.

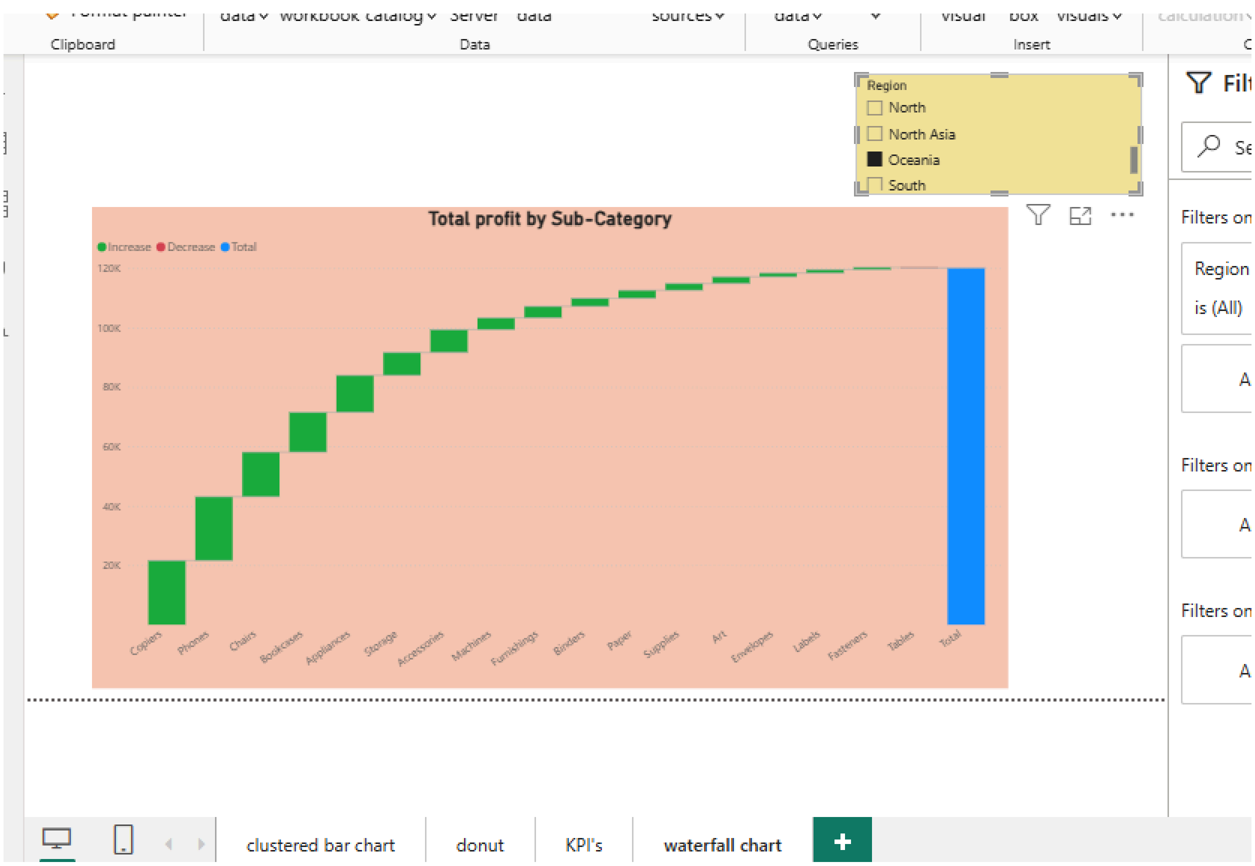
Step 2: Create a Waterfall Chart

1. In Power BI Desktop, go to the **Visualizations** pane.
2. Select the **Waterfall chart** icon.
3. Drag `Sub-Category` to the **Category** field.
4. Drag the **Total Profit** measure to the **Y-axis (Values)** field.
5. Optional: Sort by Total Profit descending to identify top/bottom performers quickly.

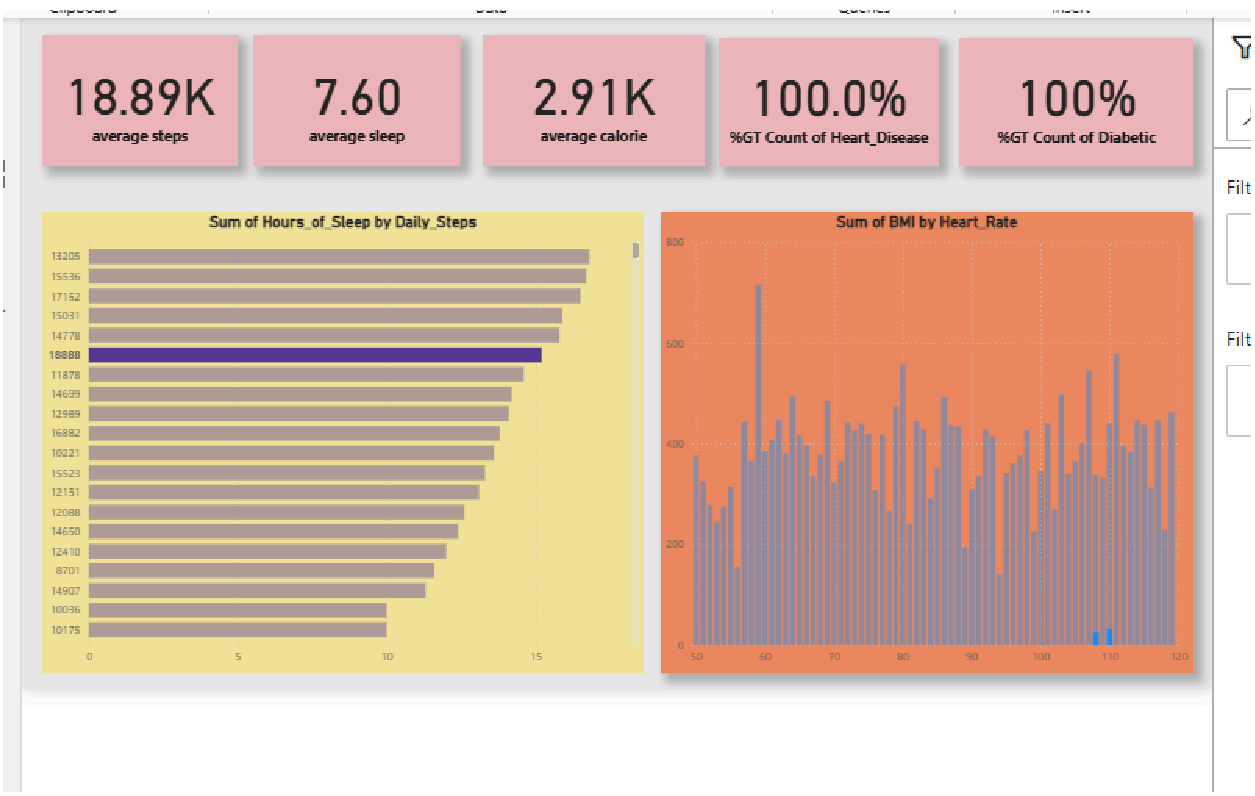
Step 3: Add a Slicer for Region

1. Select the **Slicer** visual from the Visualizations pane.
2. Drag the `Region` field into the slicer.

3. Now, selecting a region will dynamically filter the waterfall chart.



10.Scenario: VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals. The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks. Your Task: Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status): Build a one-page Power BI dashboard that answers: 1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories) 2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk? 3. Is there any visible relationship between Sleep and Physical Activity? 4. How does BMI vary across Age Groups and Genders? 5. What is the impact of smoking and alcohol on heart rate and blood pressure? 6. Segment people based on their health activity to suggest lifestyle changes


18.89K
average steps
7.60
average sleep
2.91K
average calorie
100.0%
%GT Count of Heart_Disease
100%
%GT Count of Diabetic
Sum of Hours_of_Sleep by Daily_Steps
13205
15536
17152
15031
14778
18888
11878
14699
12989
16882
10221
15523
12151
12088
14650
12410
8701
14907
10036
10175
0
5
10
15
Sum of BMI by Heart_Rate
800
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